Root finders against tower F4, signature F5 and the F6 analogue on prime curves

Run report, ecbench, 2026-10-18 22:20 UTC onward (run run-20261018T2220Z)

1. What ran

- **Protocol.** research/pkm_tower_rootfind_20261018/PROTOCOL.md, committed before any measured run.
- **Amendment 1.** It adds the AGENTS.md §5a pricing band and records the P8 rerun.
- **Code.** Commit ebed0cc51, with one ecbench binary in every session.
- **Host.** A cloud VM, isolation L1 or L2 as recorded.
- **P8.** A worker restart killed the first P8 after 20 of 36 runs. It is kept in interrupted/, and P8 was rerun in full.
- **Base.** Every IC arm used prime-kummer:t=T,g=G with G = 37, 684, 7047 and 2551. That is the 2026-10-11 round's base point for point (points_sha256).
- **Targets.** The same target seed, 202610112301, so the same targets.
- **Arms.**
  - Gröbner: f4 (tower F4), sig-f5 (signature F5) and f6 (the F6 analogue), all through pkm-tower.
  - No-algebra control: enum.
  - Root finders: rootfind-subtract:m=2 and rootfind-mitm:m=2.
  - Context: rootfind-mitm:m=3, rootfind-pairs:m=3 and rho.negation.

spec	session	curve	targets	verified	replay
P5	ECBS1h2306d857d0af	icv1-fp9-t25-33fa1ca2	8	72/72	ok
P6	ECBS1hf6c3a80d1387	icv1-fp11-t43-4d73e7f7	8	72/72	ok
P7	ECBS1hede9211000ad	icv1-fp13-tm41-756f569a	8	72/72	ok
P8	ECBS1h3c465cff7dc5	icv1-fp15-t181-50ff72d2	4	36/36	ok

- **Instrument check.** f4, sig-f5 and f6 reproduce the 2026-10-11 round's counts on 28 of 28 targets.
- **Sources.** Raw records are in sessions/<spec>/records.jsonl. Derived values are in ANALYSIS.json, written by examples/pkm_rootfind_analyze.rs.

2. Measured values (medians over 8 targets, 4 at P8)

Units.

- S: charged GAE over √r.
- M: F_p multiplications, derived as 3 × additions + 7 × doublings plus the solver's multiply-adds.
- Inversions: one per addition or doubling, plus native batched inversions.

cell	F4 S	F5 S	subtract S	mitm S	F4 M	F5 M	subtract M	mitm M
P5	25.84	25.84	29.02	41.53	8.05e7	1.82e7	2,190	2,762
P6	15.55	15.55	36.03	35.74	2.35e9	6.51e8	5,200	5,192
P7	9.36	9.36	35.72	58.08	4.07e10	1.25e10	11,039	16,882
P8	7.48	7.48	49.97	126.25	4.86e11	1.59e11	22,029	52,947

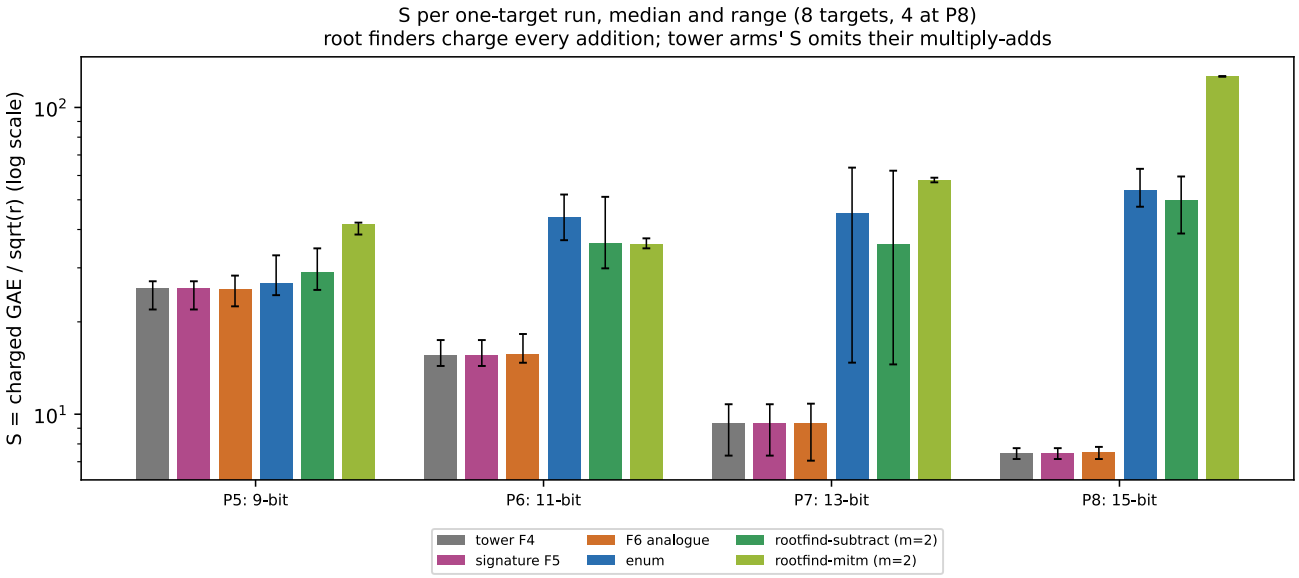


Figure 1: S per arm and cell, median and range, sessions P5–P8. The root finders charge every addition. The tower arms' S omits their multiply-adds.

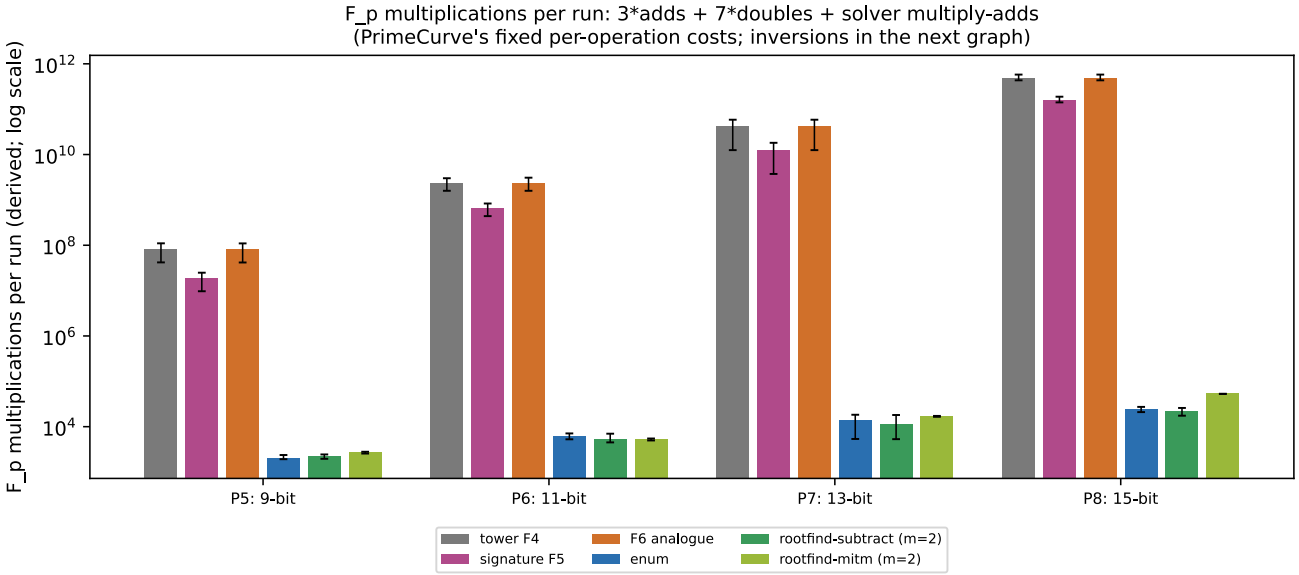


Figure 2: Derived F_p multiplications per run. F4 and F5 are dominated by solver multiply-adds; enum and the root finders have none.

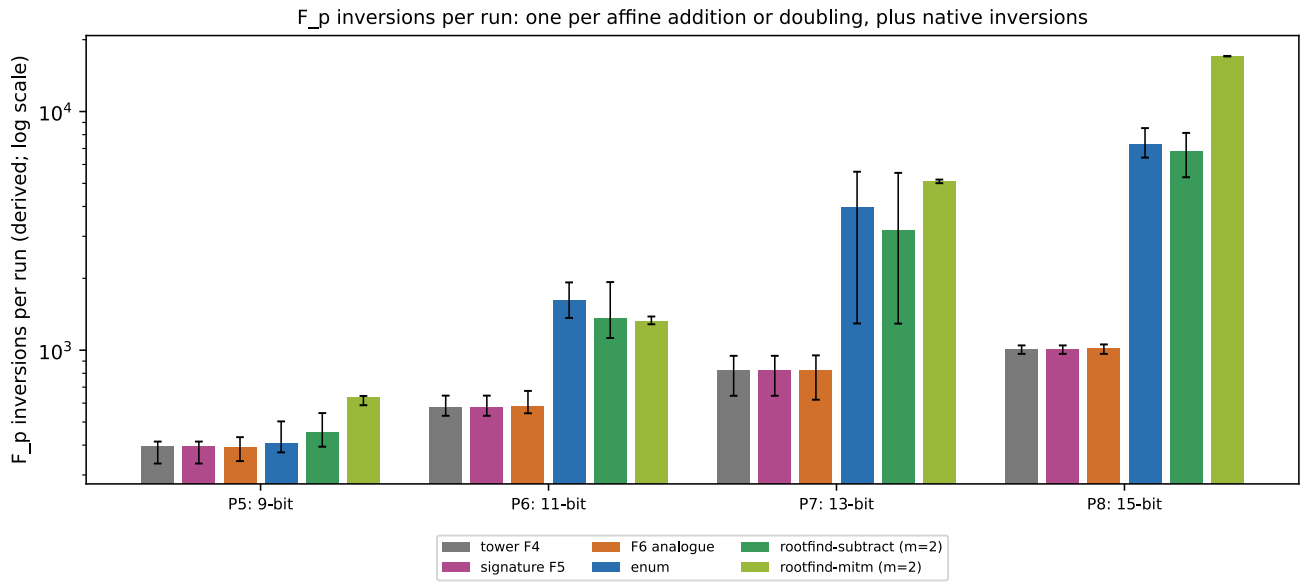


Figure 3: Derived `F_p` inversions per run. The root finders do 1.15–17 times F4′s, one per affine addition.

3. The protocol's questions against F4 and F5 (derived; rules as written)

- **Q1, `S(root)/S(F4)`** (identical against F5).
  - `rootfind-subtract`: 1.16, 2.23, 3.53, 6.70.
  - `rootfind-mitm`: 1.62, 2.29, 6.24, 16.9.
  - Rule: **not decided by S** in every cell. F4′s `S` is a lower bound.
- **Q2, `M(root)/M(F4)`.**
  - `rootfind-subtract`: 2.8e-5, 2.3e-6, 3.3e-7, 4.6e-8.
  - Against F5: 1.2e-4, 8.3e-6, 1.1e-6, 1.4e-7.
  - The root finders have more inversions on every target, so the rule reads **mixed**.
  - For F4 to be cheaper, an inversion would have to cost 1.3e6 (P5) to 8.4e7 (P8) multiplications.
- **§5a band** (`S_w`, w = 0, 0.1, 1 GAE per unpriced operation).
  - At w = 0 the root finders are above F4/F5. At 0.1 and 1 they are below by 4–8 orders of magnitude.
  - Verdict: **unresolved inside the band**, in every cell.
  - Derived break-even weight, `rootfind-subtract` against F4: 8.5e-7, 3.6e-7, 6.9e-8, 1.2e-8 GAE.
- **Relation-phase wall, root / F4:** 1e-4 (P5) down to 1e-6 (P8).
- **Context.**
  - S/rho: `rootfind-subtract` 5.4–29.2, F4 3.9–5.4 (lower bound).
  - The `m = 3` root finders cost 1.29–1.90 times `rootfind-mitm` at `m = 2`.

4. Limits

- **Derived quantities.** `M` and the inversions use the curve's fixed per-operation costs.
- **Practicality notes.** Wall time comes from mixed L1/L2 runs.
- **Scope.** No arm is below rho. There is no exponent, from four sizes up to 15 bits. The head-to-head is at `m = 2` only.
- **Interpretation.** It is in the study's `RESULT.md`.